THE MASTERMIND COACHING CLASSES

CBSE / RBSE · Nursery – Class XII · Road No. 5, VKIA, Jaipur · 9887258679

ANSWER KEY & MARKING SCHEME — MATHS · CLASS 9 · SET 01

Max. Marks: 80 · The Mastermind Coaching Classes, Jaipur · Alternate correct methods to be given full credit

SECTION A — Objective Type — 20 questions · 20 marks

Q1–Q18 are multiple-choice questions (1 mark each). Q19–Q20 are Assertion (A) and Reason (R) questions: select the correct option from (a), (b), (c), (d) given below each.

1. The point (−5, 0) lies on the:

(a) y-axis (b) x-axis

(c) third quadrant (d) second quadrant

Ans: (b) Ordinate is 0, so the point lies on the x-axis.

2. Which of the following can be the coordinates of a point in the fourth quadrant?

(a) (3, −2) (b) (−3, −2) (c) (−3, 2) (d) (3, 2)

Ans: (a) Fourth quadrant: $x > 0$, $y < 0$.

3. The midpoint of the line segment joining A(2, −6) and B(−4, 10) is:

(a) (1, 2) (b) (−1, −2) (c) (−1, 2) (d) (−3, 8)

Ans: (c) $\left(\frac{2-4}{2}, \frac{-6+10}{2}\right) = (-1, 2)$.

4. A rectangle has length (2x + 1) cm and breadth (x − 3) cm. Its perimeter (in cm) is:

(a) 6x + 4 (b) 3x − 2 (c) 3x + 4 (d) 6x − 4

Ans: (d) $P = 2[(2x + 1) + (x − 3)] = 2(3x − 2) = 6x − 4$.

5. The value of the polynomial $p(x) = 3x^2 − 4x + 1$ at $x = −1$ is:

(a) 8 (b) 0 (c) −6 (d) 2

Ans: (a) $p(−1) = 3 + 4 + 1 = 8$.

6. A cab company charges ₹40 as fixed charge and ₹14 per km. The fare (in ₹) for d km is:

(a) 40d + 14 (b) 14d + 40 (c) 54d (d) 14d − 40

Ans: (b) Fare = 14 × d + 40.

7. The absolute value of the rational number $\frac{-7}{9}$ is:

(a) $\frac{-7}{9}$ (b) $\frac{-9}{7}$ (c) $\frac{7}{9}$ (d) $\frac{9}{7}$

Ans: (c) $|x|$ is never negative: $|−7/9| = 7/9$.

8. Which of the following is an irrational number?

(a) 5.0505…(0.05 repeating) (b) √0.04

(c) √18 (d) √(81/25)

Ans: (c) $\sqrt{18} = 3\sqrt{2}$ is irrational; $\sqrt{(81/25)} = 9/5$, $\sqrt{0.04} = 0.2$ and 5.0505… (repeating) are rational.

9. Which of the following has a terminating decimal expansion?

(a) $\frac{5}{12}$ (b) $\frac{4}{15}$ (c) $\frac{11}{18}$ (d) $\frac{9}{80}$

Ans: (d) $80 = 2^4 \times 5$, only 2 and 5 occur in the denominator, so 9/80 terminates (= 0.1125).

10. On expanding $(3a − 2b)^2$, we get:

(a) $9a^2 - 12ab + 4b^2$ (b) $9a^2 - 6ab + 4b^2$
(c) $9a^2 - 12ab - 4b^2$ (d) $9a^2 + 12ab + 4b^2$

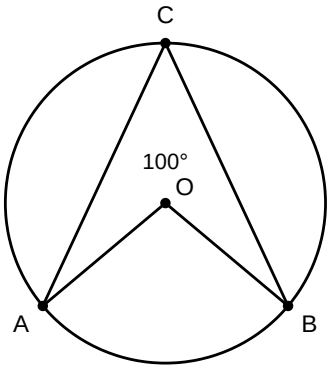
Ans: (a) $(x - y)^2 = x^2 - 2xy + y^2$.

11. Using a suitable identity, the value of 98×102 is:

(a) 9996 (b) 10004 (c) 9998 (d) 9904

Ans: (a) $(100 - 2)(100 + 2) = 100^2 - 2^2 = 10000 - 4 = 9996$.

12. In the figure, O is the centre of the circle and $\angle AOB = 100°$. Then $\angle ACB$ is equal to:



(a) 200° (b) 100° (c) 50° (d) 25°

Ans: (c) The angle at the centre is twice the angle at the remaining circumference: $\angle ACB = 100° \div 2 = 50°$.

13. A chord of length 16 cm is drawn in a circle of radius 10 cm. Its distance from the centre is:

(a) $\sqrt{164}$ cm (b) 4 cm (c) 6 cm (d) 8 cm

Ans: (c) $d^2 = 10^2 - 8^2 = 36$, so $d = 6$ cm.

14. The sides of a triangle are in the ratio 4 : 5 : 6 and its perimeter is 90 cm. The shortest side is:

(a) 15 cm (b) 30 cm (c) 20 cm (d) 24 cm

Ans: (d) One part = $90 \div 15 = 6$ cm; shortest side = $4 \times 6 = 24$ cm.

15. The area of a triangle with sides 5 cm, 12 cm and 13 cm, found using Heron's formula, is:

(a) 78 cm² (b) 32.5 cm² (c) 30 cm² (d) 60 cm²

Ans: (c) $s = 15$; area $= \sqrt{(15 \times 10 \times 3 \times 2)} = \sqrt{900} = 30$ cm².

16. A letter is chosen at random from the word 'EDUCATION'. The probability that it is a vowel is:

(a) $\frac{4}{9}$ (b) $\frac{1}{2}$ (c) $\frac{5}{8}$ (d) $\frac{5}{9}$

Ans: (d) Vowels E, U, A, I, O = 5 out of 9 letters.

17. A die is thrown once. The probability of getting a number less than 5 is:

(a) $\frac{1}{3}$ (b) $\frac{5}{6}$ (c) $\frac{1}{2}$ (d) $\frac{2}{3}$

Ans: (d) Favourable outcomes 1, 2, 3, 4 → 4/6 = 2/3.

18. If the n-th term of a sequence is $t_n = n^2 - 3n$, then t_7 is:

(a) 70 (b) −28 (c) 14 (d) 28

Ans: (d) $t_7 = 49 - 21 = 28$.

19. **Assertion (A):** $\sqrt{2} + 1$ is an irrational number.
Reason (R): The sum of a rational number and an irrational number is always irrational.

Ans: (a) Both true; R directly justifies A (1 is rational, √2 irrational).

20. **Assertion (A):** For the sequence defined by $t_1 = 2$ and $t_n = t_{n-1} + 4$ ($n \geq 2$), the third term is 12.
Reason (R): A recursive rule gives every term from the term before it.

Ans: (d) $t_2 = 6$, $t_3 = 10$, so A is false; R is a true statement.

SECTION B — Very Short Answer (2 marks each) — 5 questions · 10 marks

Internal choice is provided in one question.

21. A taxi charges ₹50 for the first 3 km and ₹18 for every km after that. What is the fare for a journey of 12 km?

Extra distance = 12 − 3 = 9 km (½)
Extra charge = 9 × 18 = ₹162 (1)
Fare = 50 + 162 = **₹212** (½)

22. Without actual division, decide whether the decimal expansion of $\frac{17}{320}$ is terminating. If it is, state the number of decimal places.

320 = $2^6 \times 5$, which has only 2 and 5 as prime factors, so 17/320 terminates (1). Number of decimal places = greater of the powers of 2 and 5 = 6 (1).

OR

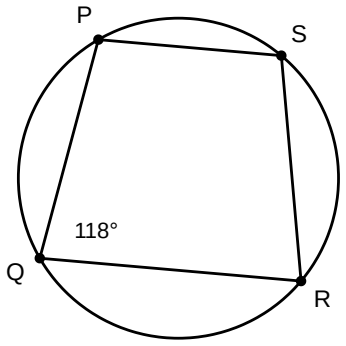
Find any two rational numbers between $\frac{1}{4}$ and $\frac{1}{3}$.

Write with common denominator 48: 1/4 = 12/48, 1/3 = 16/48 (1). Two rational numbers: **13/48 and 14/48 (= 7/24)** (1). (Any two correct numbers.)

23. Factorise: $9a^2 + 30ab + 25b^2$.

$= (3a)^2 + 2(3a)(5b) + (5b)^2$ (1) = **$(3a + 5b)^2$** $= (3a + 5b)(3a + 5b)$ (1)

24. In the figure, PQRS is a cyclic quadrilateral with ∠PQR = 118°. Find ∠PSR and state the reason.



Opposite angles of a cyclic quadrilateral are supplementary (1).
∠PSR = 180° − 118° = **62°** (1)

25. Find the second, third and fourth terms of the sequence given by $t_1 = 3$ and $t_n = 2t_{n-1} - 1$ for $n \geq 2$.

$t_2 = 2(3) - 1 = 5$; $t_3 = 2(5) - 1 = 9$; $t_4 = 2(9) - 1 = 17$. (1 mark for method, 1 for all three values)

SECTION C — Short Answer (3 marks each) — 6 questions · 18 marks

Internal choice is provided in two questions.

26. Show that $0.153846153846\ldots = \frac{2}{13}$.

Let x = 0.153846153846… (½). Six digits repeat, so multiply by 10^6: 1000000x = 153846.153846… (1).
Subtracting: 999999x = 153846 (½)
x = 153846/999999 = **2/13** (dividing numerator and denominator by 76923) (1)

27. A school canteen recorded the drink chosen by 100 students in a day:

Drink	Tea	Coffee	Juice	Milk
Students	30	20	35	15

A student is selected at random. Find the probability that the student chose (i) juice, (ii) tea or milk, (iii) not coffee.

Total = 100.
(i) P(juice) = 35/100 = **7/20** (1)
(ii) P(tea or milk) = (30 + 15)/100 = **9/20** (1)
(iii) P(not coffee) = (100 − 20)/100 = **4/5** (1)

OR

In a survey of 80 households the number of members was recorded:

Members	2	3	4	5
Households	10	25	30	15

A household is chosen at random. Find the probability that it has (i) exactly 4 members, (ii) at most 3 members, (iii) more than 4 members.

Total = 80.
(i) 30/80 = **3/8** (1) (ii) (10 + 25)/80 = **7/16** (1) (iii) 15/80 = **3/16** (1)

28. Write the coordinates of the point (i) which lies on the x-axis, 7 units to the left of the origin; (ii) which lies on the y-axis, 4 units below the origin; (iii) which is the mirror image of (−3, 5) in the x-axis.

(i) **(−7, 0)** (1) (ii) **(0, −4)** (1) (iii) reflection in the x-axis changes the sign of y: **(−3, −5)** (1)

29. Simplify: $(x + 2)^3 + (x - 2)^3$.

$(x + 2)^3 = x^3 + 6x^2 + 12x + 8$ (1)
$(x - 2)^3 = x^3 - 6x^2 + 12x - 8$ (1)
Sum = **$2x^3 + 24x$** (1)

30. A chord of length 30 cm is drawn in a circle of diameter 34 cm. Find (i) the distance of the chord from the centre; (ii) the length of another chord of this circle which is at a distance of 15 cm from the centre.

r = 17 cm. (i) Half chord = 15; $d^2 = 17^2 - 15^2 = 64$, d = **8 cm** (1½)
(ii) Half chord$^2 = 17^2 - 15^2 = 64$ → half = 8, chord = **16 cm** (1½)

31. Using Heron's formula, find the area of a triangle whose sides are 17 cm, 25 cm and 26 cm.

s = (17 + 25 + 26)/2 = 34 cm (1)
Area = √[34 × 17 × 9 × 8] (1) = √41616 = **204 cm²** (1)

OR

The sides of a triangle are in the ratio 3 : 4 : 5 and its perimeter is 72 cm. Find its area.

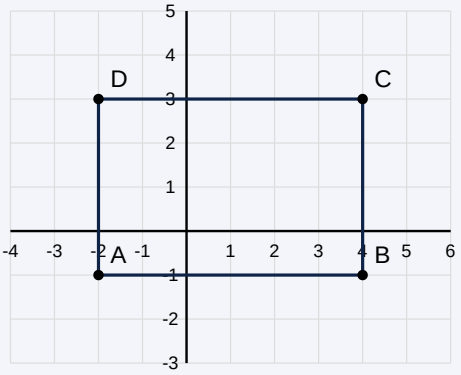
Sides: 18 cm, 24 cm, 30 cm (1); s = 36 (½)
Area = √[36 × 18 × 12 × 6] = √46656 = **216 cm²** (1½)

SECTION D — Long Answer (5 marks each) — 4 questions · 20 marks

Internal choice is provided in two questions.

32. An architect marks the four corners of a rectangular courtyard on a grid as A(−2, −1), B(4, −1), C(4, 3) and D(−2, 3).
(a) Plot the points on a coordinate plane and join them in order. (2 marks)

(b) Find the lengths AB, BC, CD and DA. (1½ marks)
(c) Using the diagonals, justify that ABCD is a rectangle but not a square. (1½ marks)



(a) Correct plotting of 4 points (1½) and joining (½): (b) AB = 6, BC = 4, CD = 6, DA = 4 units (1½)
(c) AC = $\sqrt{(6^2 + 4^2)} = \sqrt{52}$, BD = $\sqrt{(6^2 + 4^2)} = \sqrt{52}$ → equal diagonals and opposite sides equal ⇒ rectangle (1). Adjacent sides $6 \neq 4$ ⇒ not a square (½)

33. How many 3-digit numbers leave remainder 2 when divided by 6? Find the sum of all such numbers. (2½ + 2½)

Numbers: 104, 110, 116, …, 998 — an AP with a = 104, d = 6, l = 998 (1)
n = (998 − 104)/6 + 1 = **150** (1½)
Sum = n/2 (a + l) = 75 × 1102 (1) = **82650** (1½)

34. A battery company tested 500 batteries and recorded their life:

Life (hours)	0 – 50	50 – 100	100 – 150	150 – 200
Number of batteries	20	70	190	220

Find the probability that a randomly chosen battery lasts (i) less than 50 hours (1), (ii) between 50 and 150 hours (2), (iii) more than 150 hours (2).

Total = 500. (i) 20/500 = **1/25** (1) (ii) (70 + 190)/500 = **13/25** (2) (iii) 220/500 = **11/25** (2)

O R

A box contains 6 blue pens, 8 black pens and 4 red pens. A pen is drawn at random. Find the probability that it is (i) blue (1), (ii) black or red (2), (iii) neither blue nor black (2).

Total pens = 18. (i) 6/18 = **1/3** (1) (ii) (8 + 4)/18 = **2/3** (2) (iii) 4/18 = **2/9** (2)

35. The length of a rectangular play area is 5 m more than its breadth, and its area is 84 m². Form a quadratic expression in the breadth x, factorise it and find the perimeter of the play area.

Let breadth = x m, length = (x + 5) m. Area: $x(x + 5) = 84 \Rightarrow x^2 + 5x - 84 = 0$ (1½)
$x^2 + 12x - 7x - 84 = (x + 12)(x - 7) = 0$ (1½) ⇒ x = 7 (breadth cannot be negative) (1)
Length = 12 m. Perimeter = 2(12 + 7) = **38 m** (1)

O R

A carpentry student uses 3 large square tiles of side x units, 10 rectangular strips of length x and width 1 unit, and 7 small square tiles of side 1 unit to make one large rectangle without overlap. Find the length and breadth of the rectangle in terms of x.

Total area = $3x^2 + 10x + 7$ (1½)
Split the middle term: $3x^2 + 3x + 7x + 7$ (1½) = $3x(x + 1) + 7(x + 1)$ (1) = **(3x + 7)(x + 1)** (1)
Length = (3x + 7) units, breadth = (x + 1) units.

SECTION E — Case-Study Based Questions (4 marks each) — 3 questions · 12 marks

Read each case carefully and answer the questions that follow. Internal choice is provided in every 2-mark sub-part.

36. Priya's mobile data pack gives her 60 GB at the start. She uses 4 GB of data every day. The data (in GB) left after x days is given by D = 60 − 4x.

(i) How much data is left after 5 days?

> D = 60 − 20 = **40 GB**

(ii) State the degree of D = 60 − 4x and the coefficient of x.

> Degree = **1** (linear polynomial); coefficient of x = **−4** (negative ⇒ linear decay).

(iii) After how many days will only 12 GB of data be left?

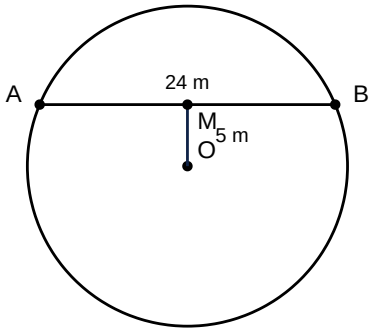
> 60 − 4x = 12 (1) ⇒ 4x = 48 ⇒ x = **12 days** (1)

OR

(iii) After how many days will her data pack be exhausted?

> 60 − 4x = 0 (1) ⇒ x = **15 days** (1)

37. A circular pond with centre O is built in a park. A fence AB, 24 m long, is fixed as a chord of the pond. The perpendicular distance of the fence from the centre, OM, is 5 m.



(i) Find the radius of the pond.

> OM ⊥ AB bisects it: AM = 12 m. $r^2 = 12^2 + 5^2 = 169$ ⇒ r = **13 m**

(ii) Another fence CD, also 24 m long, is fixed in the same pond. How far is it from the centre?

> Equal chords of a circle are equidistant from the centre ⇒ **5 m**.

(iii) A walking path PQ is a chord at a distance of 12 m from O. Find its length.

> Half of PQ = $\sqrt{(13^2 - 12^2)} = 5$ (1) ⇒ PQ = **10 m** (1)

OR

(iii) Four lamps are placed at A, B, D and E on the boundary forming a cyclic quadrilateral ABDE. If ∠ADE = 75°, find ∠ABE and give the reason.

> Opposite angles of a cyclic quadrilateral are supplementary (1) ⇒ ∠ABE = 180° − 75° = **105°** (1)

38. A residents' association has a triangular garden whose sides are 10 m, 17 m and 21 m.

(i) Find the semi-perimeter of the garden.

> s = (10 + 17 + 21)/2 = **24 m**

(ii) Find the area of the garden using Heron's formula.

> Area = $\sqrt{[24 \times 14 \times 7 \times 3]} = \sqrt{7056}$ = **84 m²**

(iii) Find the length of the altitude drawn to the longest side.

> Longest side = 21 m. ½ × 21 × h = 84 (1) ⇒ h = **8 m** (1)

O R

(iii) Grass is to be laid over the whole garden at ₹25 per m². Find the total cost.

> Cost = 84 × 25 (1) = **₹2100** (1)